

Trainings ▾

General Information

This procedure applies to rear gear train engines only. For front gear train engines, the preferred method of removing the camshaft gear is to first remove the camshaft. Refer to procedure 001-008 (/qs3/pubsys2/xml/en/procedures/40/40-001-008-tr.html). Once the camshaft is removed, the camshaft gear can be removed. Refer to Procedure 001-013 (/qs3/pubsys2/xml/en/procedures/40/40-001-013.html).

Camshaft gear removal/installation tool, Part Number 3823589, can be used to replace the camshaft gear with the camshaft partially installed. This method should **only** be applied to engines used in applications that prohibit the removal of the camshaft in chassis.

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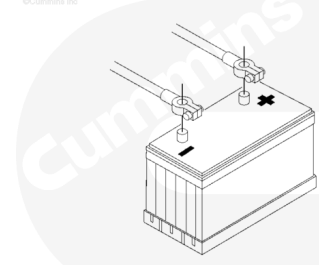
Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

Disconnect the batteries.

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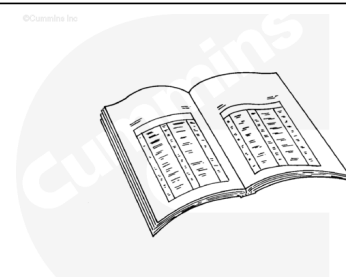


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⚠ WARNING ⚠

Support the rear of the engine using the rear support attached to the rear of the cylinder block. Failure to support the engine can cause serious personal injury.

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ck800wa

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.

- Remove the transmission. Refer to the original equipment manufacturer's instructions
- Remove the starting motor. Refer to Procedure 013-020 (</qs3/pubsys2/xml/en/procedures/40/40-013-020-tr.html>)
- Remove the flywheel or flexplate. Refer to Procedure 016-005 (</qs3/pubsys2/xml/en/procedures/40/40-016-005-tr.html>) or Procedure 016-004 (</qs3/pubsys2/xml/en/procedures/40/40-016-004-tr.html>)
- Remove the flywheel housing. Refer to Procedure 016-006 (</qs3/pubsys2/xml/en/procedures/40/40-016-006-tr.html>)
- Lock the fuel pump. Refer to Procedure 005-014 (</qs3/pubsys2/xml/en/procedures/40/40-005-014-tr.html>), Fuel Injection Pump, Rotary, or Procedure 005-012 (</qs3/pubsys2/xml/en/procedures/40/40-005-012-tr.html>), Fuel Injection Pumps, In-Line.

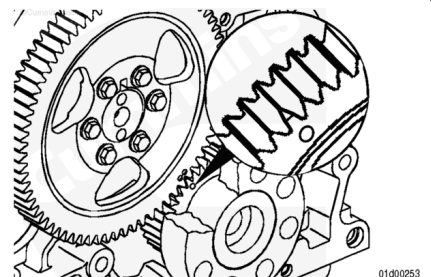
Note : Failure to lock the fuel pump may result in improper fuel pump timing during reassembly.

Remove

Rotate the engine to allow the camshaft timing marks to align.

Note : The engine can have a mark on the camshaft gear or chamfered tooth.

Service Tip: Engines equipped with air compressors may require that the air compressor be timed to the engine. To ensure that the air compressor is properly timed when the camshaft gear is later installed, scribe an alignment line on the air compressor and camshaft gear before removing the camshaft gear.



On engines equipped with an air compressor/accessory drive, it may be necessary to loosen/remove some the air compressor/accessory drive mounting hardware in order to

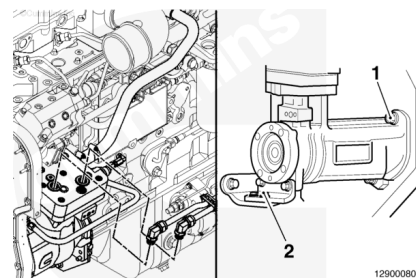
remove the camshaft gear. It is **not** necessary to remove the air compressor/accessory drive completely.

Loosening/remove some of the air compressor/accessory drive mounting hardware will give enough clearance to remove the camshaft gear.

Loosen the air compressor mounting fasteners (1).

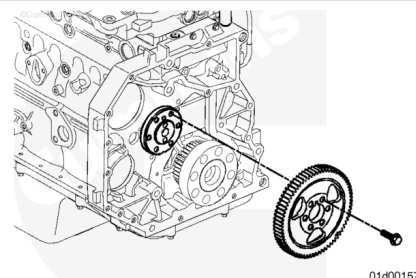
Remove the two capscrew securing the air compressor support (2).

If equipped with a hydraulic pump driven off of the air compressor, it may be necessary to remove and/or loosen some or all of the mounting fasteners. Refer to the OEM Instructions.



Remove the camshaft gear capscrews.

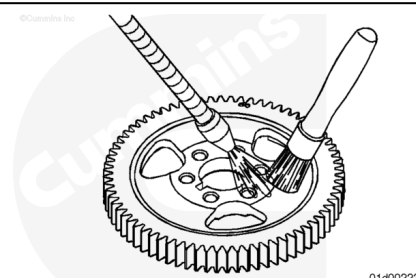
Remove the camshaft gear.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

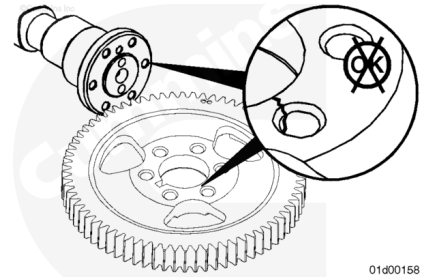


⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

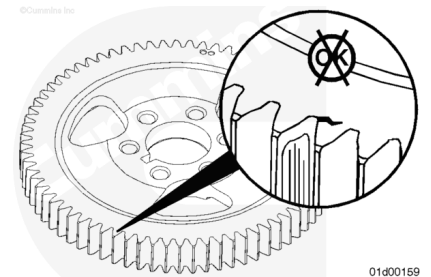
Clean the camshaft gear. Dry with compressed air.

Inspect for cracks and fretting at the capscrew holes on the camshaft and camshaft gear.



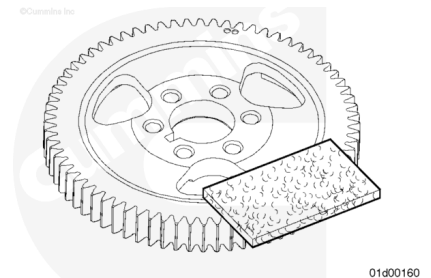
Inspect for fretting on the camshaft gear teeth.

Inspect for cracked or broken teeth on the camshaft gear.



Remove any frets, burrs, or raised metal with abrasive pad, Part Number 3823258.

If frets, burrs, or raised metal can **not** be removed with abrasive pad, Part Number 3823258, replace the gear.



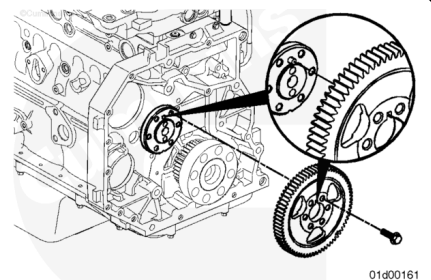
Install

⚠ CAUTION ⚠

To reduce the possibility of engine damage, make sure the camshaft rotates freely.

Align the camshaft gear with the pin in the camshaft and the mark on the crankshaft gear.

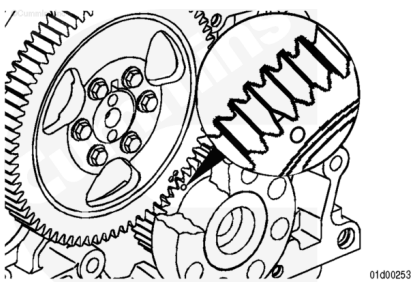
Install camshaft gear onto the camshaft.



Check to make sure the timing marks on the camshaft gear align with the timing marks on the crankshaft gear. If equipped with an air compressor, make sure to align the line that was scribed on the camshaft gear and air compressor gear during

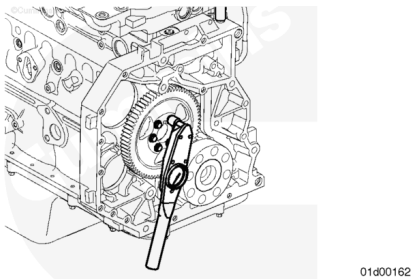
the camshaft gear removal step. If this was **not** done, refer to Procedure 012-014 (/qs3/pubsys2/xml/en/procedures/40/40-012-014-tr.html) for air compressor timing procedures.

Note : The engine can have a mark on the crankshaft gear or chamfered tooth.



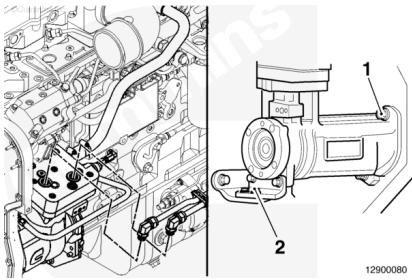
Install camshaft gear capscrews and tighten.

Torque Value: 36 n•m [27 ft-lb]



⚠ CAUTION ⚠

On engines equipped with an air compressor/accessory drive, before tightening the air compressor mounting fasteners, inspect to see that the seals/gaskets are in place. Failure to inspect may result in an oil leak or blockage of the air compressor oil passages, which will damage the air compressor.



Tighten the air compressor/accessory drive mounting fasteners (1).

Torque Value: 77 n•m [57 ft-lb]

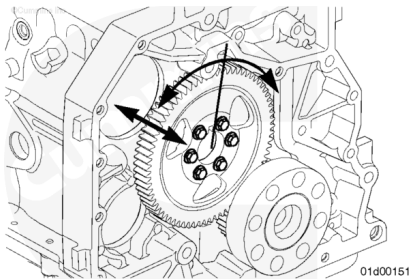
Install the two capscrew securing the air compressor support (2).

M8	24 n.m	[18 ft-lb]
M10	43 n.m	[32 ft-lb]
M12	77 n.m	[57 ft-lb]

If equipped with a hydraulic pump driven off of the air compressor, install and/or tight any of the mounting fasteners removed. Refer to the OEM Instructions.

Measure

Using gauge, Part Number 3824564, and magnetic base, Part Number 3377399, check the camshaft end play.



mm		in
0.100	MIN	0.004
0.360	MAX	0.014

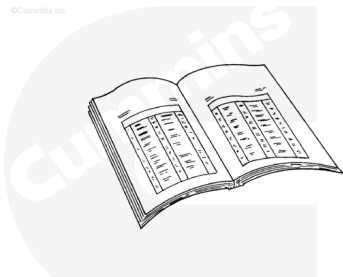
Check the camshaft backlash.

mm		in
0.076	MIN	0.003
0.280	MAX	0.011

Finishing Steps

⚠ WARNING ⚠

Support the rear of the engine using the rear support attached to the rear of the cylinder block. Failure to support the engine can cause serious personal injury.



⚠ WARNING ⚠

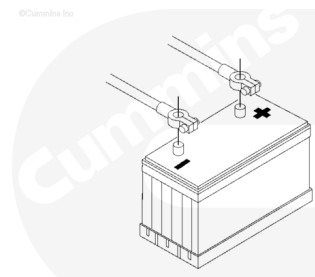
This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.

- Unlock the fuel pump. Refer to Procedure 005-014 (/qs3/pubsys2/xml/en/procedures/40/40-005-014-tr.html), Fuel Injection Pump, Rotary, or Procedure 005-012 (/qs3/pubsys2/xml/en/procedures/40/40-005-012-tr.html), Fuel Injection Pumps, In-Line
- Install the flywheel or flexplate. Refer to Procedure 016-005 (/qs3/pubsys2/xml/en/procedures/40/40-016-005-tr.html) or Procedure 016-004 (/qs3/pubsys2/xml/en/procedures/40/40-016-004-tr.html)
- Install the flywheel housing. Refer to Procedure 016-006 (/qs3/pubsys2/xml/en/procedures/40/40-016-006-tr.html)

- Install the starter motor. Refer to Procedure 013-020 (/qs3/pubsys2/xml/en/procedures/40/40-013-020-tr.html)
- Install the transmission. Refer to the original equipment manufacturer's instructions.

⚠ WARNING ⚠

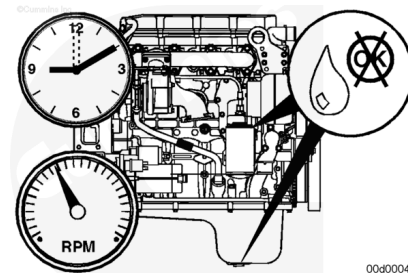
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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Reconnect the batteries.

Operate engine and check for leaks.



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